GITHUB : https://github.com/MDAzhar7799/Linux-File-Monitoring-Auto-Backup1

Submission of Project Title with Synopsis

Synopsis Contents (List Only)

1. **Title of the Project: "Linux File Monitoring & Auto-Backup System Using Shell Scripting"**
2. **Introduction:** Linux is one of the most powerful and secure operating systems widely used in servers, data centres, and enterprise environments. System administrators often face challenges in tracking changes in important files or directories.
   This project aims to build an automated file monitoring and backup system using Linux shell scripting.
   The system checks a folder for updates and automatically creates a backup whenever a change occurs.

   This improves system reliability, reduces human error, and is extremely useful for developers, administrators, and network engineers.

3. **Objectives of the Project:**

   - To understand Linux file management and process automation.
   - To create an automated script that monitors a directory.
   - To generate auto-backups whenever files are added, modified, or deleted.
   - To practice shell scripting, cron jobs, and file handling commands.
   - To reduce manual work and improve efficiency for system administrators.

4. **Problem Statement / Need of the Project:**

   - In enterprises, many critical files are frequently updated (config files, logs, documents).
     Manually checking changes and taking backups is:

   ❌ Time-consuming
   ❌ Error-prone
   ❌ Not reliable when multiple admins work on the same system

   - A lightweight **automated backup and monitoring tool** is needed.
   - This project solves the problem by:

✓ Monitoring files in real-time
✓ Automatically creating backups
✓ Logging all changes
✓ Reducing the workload of system administrators

5. **Scope of the Project:**

This project can be used in:

- Linux servers
- Student labs
- Software development environments
- Web hosting/DevOps projects
- Any system needing auto-backup and monitoring

Future enhancements can include:

- Email alert system
- GUI dashboard
- Remote backup (cloud/SCP)
- Encryption for security

6. **Methodology / Tools Used:**

❖ **Tools Used**

- Linux OS (Ubuntu, Kali, CentOS, Debian, etc.)
- Bash Shell
- Cron Job Scheduler
- Linux Commands: inotifywait, resync, cp, mv, tar
- Log files for change tracking

❖ **Methodology**

- Identify the directory to monitor.
- Write a shell script using inotify-tools or cron to detect file changes.
- Create logic to take an automatic backup when a change is detected.
- Store backups with timestamps.
- Maintain a log file containing change history.
- Test the system with file additions, modifications, and deletions.

- Schedule automatic runs using cron if needed.
- Evaluate performance and correctness.

7. **Expected Outcomes:**

By the end of this project, the student will be able to:

- Build a complete automation tool in Linux.
- Write effective shell scripts for real system use-cases.
- Monitor directories in real-time.
- Automatically back up important data.
- Understand process scheduling and logging in Linux.
- Improve knowledge about Linux environment and scripting.

The project will produce:

✓ A working shell script
✓ Automatic backup folder
✓ Log file showing detected changes
✓ Complete documentation (this file)

8. **Project Timeline:**

Week 1	Understanding Linux scripting basics, deciding folder structure
Week 2	Write file monitoring script using shell commands
Week 3	Implement automatic backup creation and timestamps
Week 4	Add logging + cron job scheduling
Week 5	Testing with sample files and refinement
Week 6	Documentation, EDA, preprocessing, final submission

9. **References**

Front page format is : SYNOPSIS

Project Title: “Linux File Monitoring & Auto-Backup System Using Shell Scripting”

Submitted By:
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Course Code: INT331
Course Title: FUNDAMENTALS OF DEVOPS

Department: CSE B.Tech

Submitted To:
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Session: KO023